Exp. 4: Age Scheduling in Networks

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Overview

1. Introduction
2. System Model
3. Problem Statement
4. Expected Outcomes

Old News is No News

- With coming of 6G, we are going for ultra-fast transmission and massive data exchange.
- Yet, beyond speed and volume lies a fundamental question: *how fresh is the information being delivered?*
- Because in modern networks, stale information can be as harmful as no information at all.

Information freshness in Networks:

How much up-to-date information we have with respect to the source.

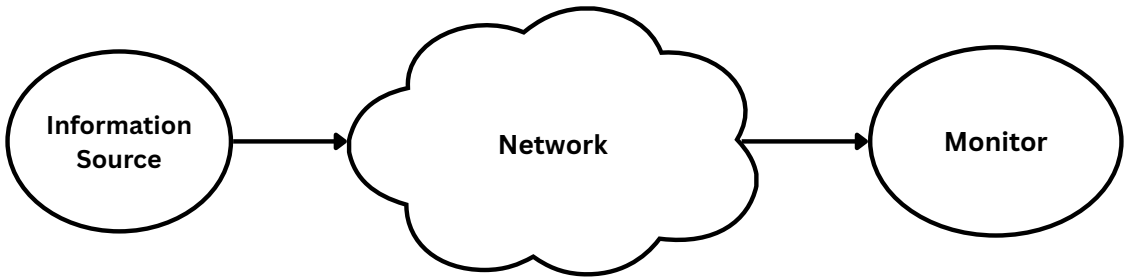


Figure: Source Monitor pair

- Fresh information is critical for:
  - Networked control systems
  - Distributed monitoring systems
  - Autonomous and IoT applications

Example: Waymo

- Autonomous Self driving cars by Google.
- uses Lidars, Radars and Cameras, all networked together to map out the surroundings to navigate the environment.
- Car needs to know the exact position of a pedestrian now, not where they were 500ms ago.



Age Of Information(AOI)

Definition: *If the monitor's latest received update was generated at time $u(t)$, then the AoI at time t is:*

$$\Delta_M(t) = t - u(t)$$

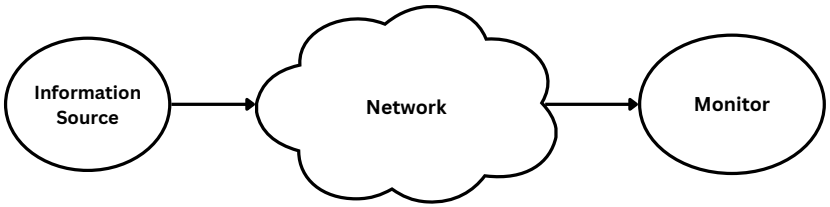


Figure: Source Monitor pair

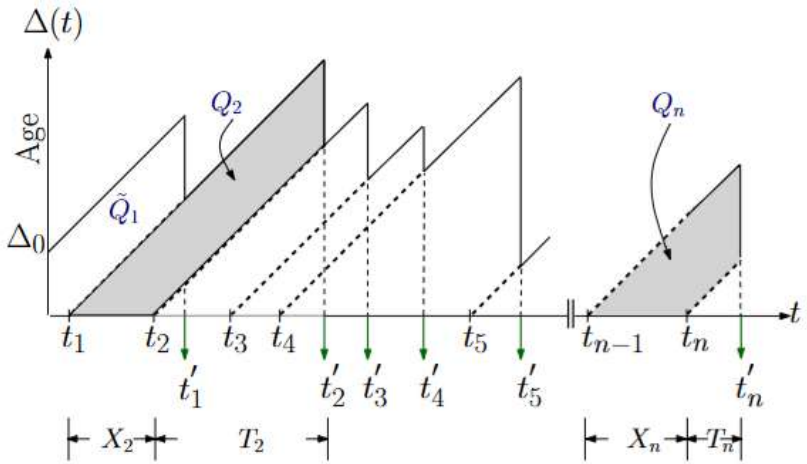


Figure: Age Evolution at Monitor

System Model

We will consider our Network as a Queue-server problem.

- Source sends update with an arrival rate λ.
- Queue has a server with service rate μ and a Buffer size B.
- Monitor wants to track the Source as timely as possible with AOI as a metric.

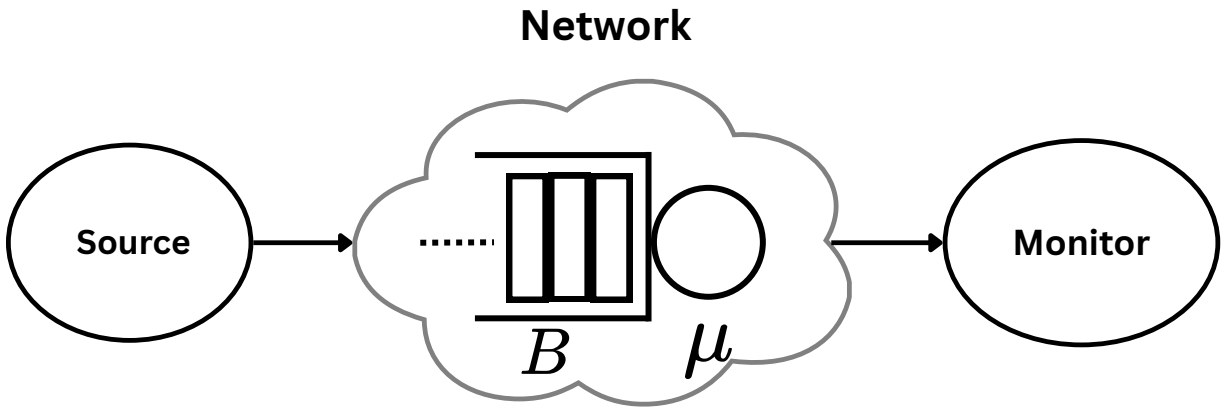


Figure: Network as Queue

Scheduling Problem

Scheduling under different policies in the queue.

1. **First Come First Serve (FCFS)**
   Update packet which arrives first in the queue gets served first.
2. **Last Come First Serve (LCFS)**
   Update packet which arrives last in the queue gets served first.
3. **Random Scheduling**
   Out of all update packets present in the queue, one is chosen randomly for service.
4. **Round Robin**
   The server gives a fixed time t to each update packet in the queue. If the service is not completed within that time, the packet goes to the end of the queue and waits for its turn to get service again.

Problem Statement

Primary Tasks:

1. Model a Network as a Queue-Server in **matlab** where there a Source generates and sends packets and the server provides service to these packets as a random process (exponentially distributed).
2. Find Average Age at the monitor under the queue Scheduling policies:
   1. FCFS
   2. LCFS
   3. Random Scheduling

   and plot the Age(Δ_M) vs Load(ρ) plot for each scheduling policy. and comment about your observations in each plots and their comparison.

Secondary Tasks:

1. Do Queue scheduling for Round Robin.
2. Generate a Sawtooth Age plot (under FCFS).

Expected Outcomes

Example:

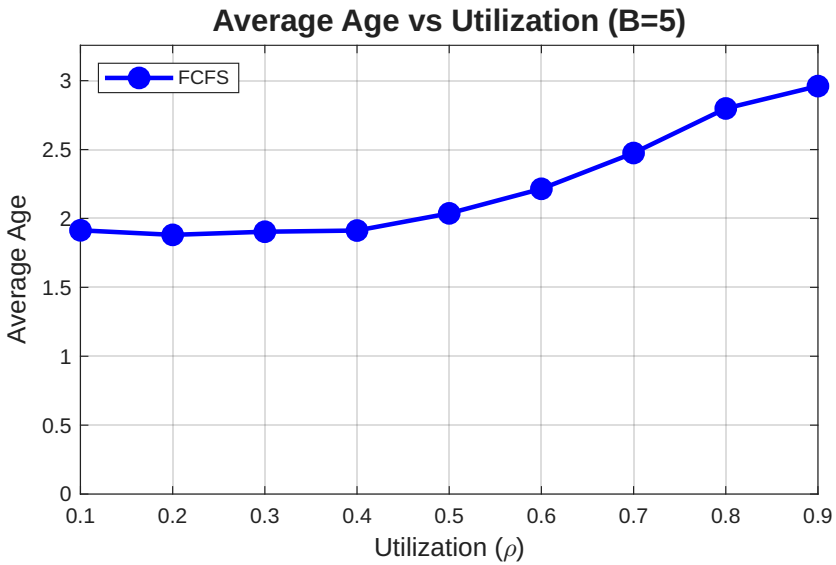


Figure: under FCFS scheduling

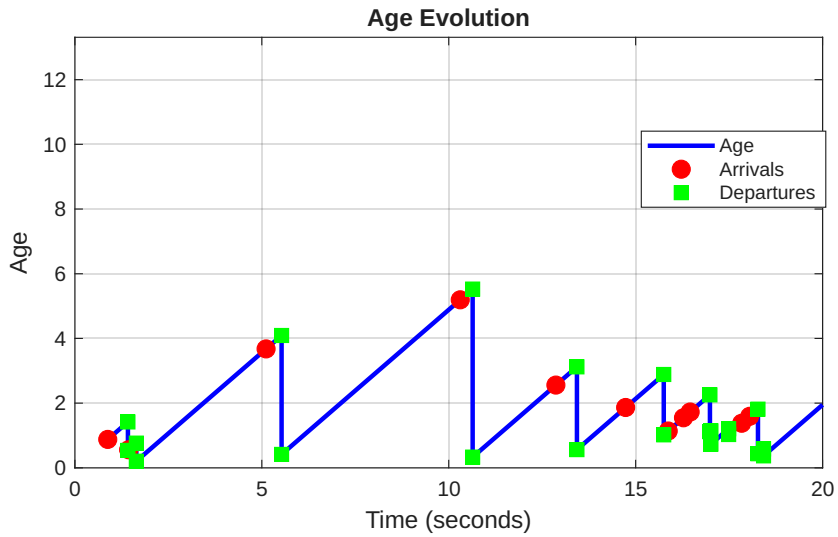


Figure: Sawtooth plot

References

1. S. Kaul, R. Yates, and M. Gruteser, “Real-time status: How often should one update?” IEEE INFOCOM, 2012.
2. Yates, Roy D., et al. ”Age of information: An introduction and survey.” IEEE Journal on Selected Areas in Communications(2021)
3. Kaswan, Priyanka, et al. ”Age of Information in Gossip Networks: A Friendly Introduction and Literature Survey Invited Paper.” IEEE Transactions on Communications (2025).